

E14 Calculating residence time distribution quantities

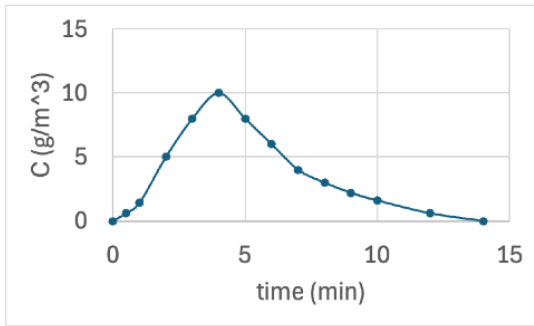
This is Fogler Examples 16-1 and 16-2.

A sample of the tracer hytane was injected as a pulse into a reactor and the exit concentration was measured as a function of time. The data is given below.

1. Construct a curve showing $C(t)$.
2. Construct the $E(t)$ curve.
3. Construct the $F(t)$ curve.
4. Determine the mean residence time.
5. Calculate the fraction of fluid that spends between 3 and 6 minutes in the reactor.

t (min)	0	0.5	1	2	3	4	5	6	7	8	9	10	12	14
C (g/m ³)	0	0.6	1.4	5	8	10	8	6	4	3	2.2	1.6	0.6	0

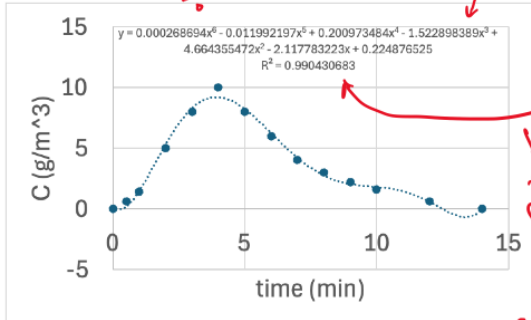
①



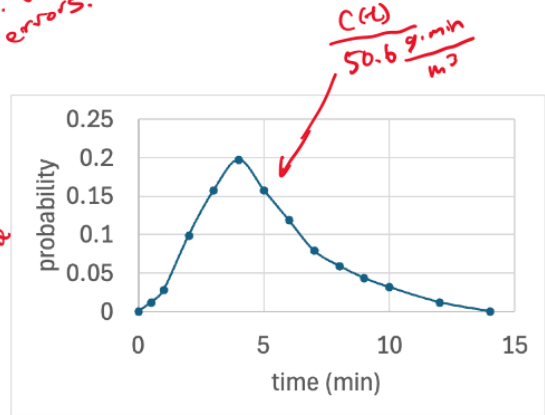
here we just plot the data in the table

② $E(t) = \frac{C(t)}{\int_0^\infty C(t) dt}$

integrals aren't sensitive to "overflow" polynomials. Use a lot of sig figs to avoid errors.

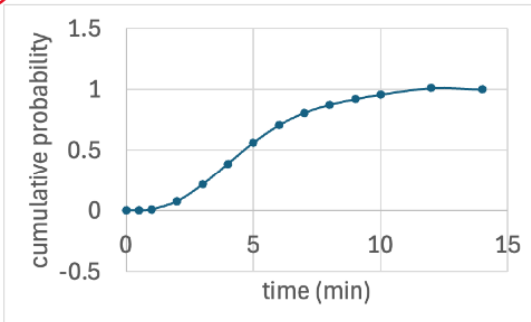


integrate from 0 to 14 to approximate $\int_0^\infty C(t) dt$



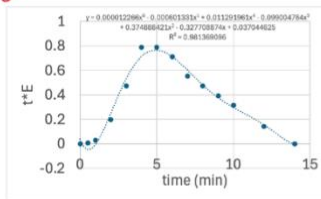
$\int_0^{14} y dt \approx \int_0^\infty C(t) dt = 50.6 \frac{g \cdot min}{m^3}$

③



$F(t) = \int_0^t E(t) dt$.
this plot is $\int_0^t y dt$ for each t value in the table

④



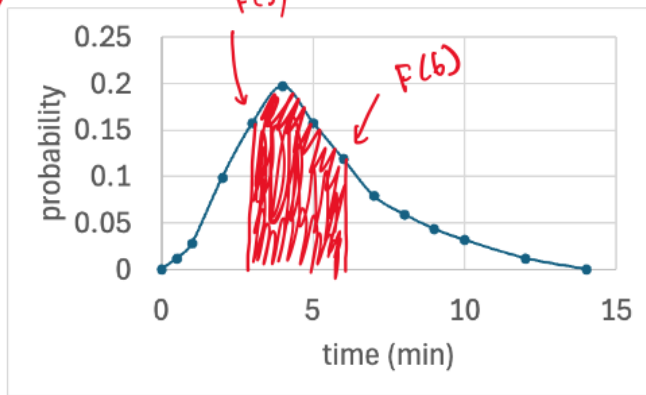
$t_m = \int_0^\infty t \cdot E(t) dt$

new data table

t	0	0.5	1	2	3	4	5	6	7	8	9	10	12	14
t · E(t)	0	0.006	0.029	0.20	0.47	0.77	0.77	0.71	0.65	0.47	0.39	0.32	0.19	0

plot new data, fit using lots of sig figs in your coefficients, $\int_0^{14} y dt$. Get $t_m = 5.2$ min

5



integrate $E(t)$ between
3 & 6 minutes to
get fraction.

$$\int_3^6 y dt = F(6) - F(3) = 0.494$$

So 49% of molecules
have τ between
3 & 6 minutes